Machine Learning — Challenge 2
Learning with Limited Labels: A Semi-Supervised Approach to Public Data

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Overview

This challenge introduces `semi-supervised learning` (*SSL*) for undergraduate students using real-world open government data from U.S. sources. Students will work with datasets that contain a small labeled subset and a large unlabeled pool. The goal is to compare a supervised baseline against one or more semi-supervised methods and to analyse when and why SSL helps (or hurts).

Learning Objectives

Students will learn to:

- Implement reproducible *supervised baselines* (`Logistic Regression` / `Random Forest`).
- Apply a semi-supervised algorithm (`self-training` / `label propagation` / `co-training`).
- Design *controlled experiments* with limited labels and report statistical variation.
- Diagnose failure modes (*bad pseudo-labels*, *class imbalance*, *leakage*) and improve preprocessing.

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Core idea and constraints

Provide each group with:

- One moderately-sized real *dataset* drawn from a *U.S. open government* source (tabular or tabular+time).
- A small labeled subset (10−−20% of the available labeled data) and a larger unlabeled pool. Labels for the unlabeled pool are held out and used only for evaluation.

Students must:

1. Train a `supervised baseline` using only the labeled subset.
2. Apply a `semi-supervised method` and report the method details (*thresholds*, *selection rules*, *iterations*).
3. Compare performance on a held-out test set and provide analysis explaining the observed results.

Required Techniques

Baselines (required): `Logistic Regression` and `Random Forest` (*scikit-learn*).

Semi-supervised (choose at least one):

- `Self-training / Pseudo-labeling` (recommended; easiest to implement and educational).
- `Label propagation / Label spreading` (available in *scikit-learn*).
- `Co-training` (optional; for stronger students who want to experiment with multiple feature views).

Suggested Workflow

1. *Data exploration* and *preprocessing* (missing values, feature selection/engineering, class balance).
2. *Partition data*: train-labeled (10 − −20%), unlabeled pool, validation (optional), and held-out test set.
3. Train *supervised baselines* on the labeled subset.
4. Apply `SSL`:
   a. *Predict labels* for the unlabeled pool.
   b. *Select high-confidence predictions* (use probability thresholds or top-k per class).
   c. *Retrain model with expanded labeled set*; iterate if desired.
5. Evaluate on the held-out test set using appropriate `metrics` (*accuracy*, *precision/recall/F1*, *AUC* when applicable) and report per-class performance and variance across seeds.

Experimental Design Recommendations

- Repeat each experimental condition with at least three random seeds; report mean and standard deviation.
- Use strict separation: never use held-out test labels during SSL selection.
- Control pseudo-labeling growth (limit number per iteration, require high confidence) to avoid error propagation.
- Handle class imbalance explicitly (stratified splits, class-weighting, per-class confidence thresholds).

Evaluation Criteria

Grading will consider:

- Improvement over baseline (quantitative) 30%.
- Correct and well-justified use of SSL techniques 20%.
- Experimental design and reproducibility (seeds, environment, instructions) 15%.
- Interpretation of results and discussion of failure modes 15%.
- Code quality, documentation & video 20%.

Common pitfalls (teaching moments)

- *Pseudo-label noise*: aggressive labeling can degrade performance.
- *Confidence misuse*: probabilities are not calibrated; verify before trusting thresholds.
- *Leakage*: using test labels (even indirectly) during selection or tuning.
- *Overfitting small labeled set*: prefer simple models and use cross-validation where possible.

Domains and dataset suggestions

Instead of prescribing a single dataset, groups pick a domain and one concrete dataset from `public U.S. sources` *(data.gov)*. Below each domain includes: **why** it is suited for `semi-supervised learning`, concrete **dataset sources** (`agency / dataset` name and example endpoints), characteristic modalities, and concise **project ideas** that emphasize `semi-supervised challenges`.

1. Transportation & Traffic Safety
    - Why: large, structured records with many covariates and frequent label sparsity or noisy annotations; useful to test imbalance-handling and spatial/temporal propagation of labels.

- Sources: NHTSA Fatality Analysis Reporting System (FARS) (https://www.nhtsa.gov/research-data/fatality-analysis-reporting-system), Bureau of Transportation Statistics crash datasets (https://data.bts.gov/).
- Modalities: tabular (categorical + numeric), timestamps, geolocation.
- Project ideas: predict accident severity (binary/multi-class), identify high-risk segments via semi-supervised geospatial propagation, or quantify recall improvement for minority classes using pseudo-labeling.

2. Environment & Air Quality

   - Why: sensor networks produce noisy and irregularly-sampled observations with many unlabeled intervals; SSL can exploit temporal and spatial continuity.
   - Sources: EPA Air Quality System (AQS) Data Mart (https://aqs.epa.gov/aqsweb/airdata/download_files.html), NOAA station/climate datasets (https://www.ncdc.noaa.gov/).
   - Modalities: time series, tabular sensor features, geospatial fields.
   - Project ideas: classify AQI categories from partial labels, impute/propagate air quality labels across nearby stations, anomaly detection with semi-supervised pretraining.

3. Public Health & Epidemiology

   - Why: high-impact domain with imbalanced and privacy-limited labels; realistic for semi-supervised scenarios where confirmed cases are a small subset.
   - Sources: CDC WONDER (https://wonder.cdc.gov/), BRFSS (https://www.cdc.gov/brfss/), Medicare public use files.
   - Modalities: tabular regional or patient-level aggregates, temporal trends.
   - Project ideas: classify regional risk tiers for a condition, predict readmission risk with few labeled patients, measure robustness of SSL under label noise and biased sampling.

4. Education Systems

   - Why: interpretable features and policy relevance; SSL enables leveraging large administrative records with only a subset of labeled outcomes.
   - Sources: NCES (Common Core of Data, https://nces.ed.gov/ccd/), state education open-data portals.
   - Modalities: tabular student/school attributes, categorical data.
   - Project ideas: predict student performance tiers, early-warning dropout classifiers, evaluate fairness and calibration when adding pseudo-labels across demographic groups.

5. Economics & Socioeconomic Data

   - Why: combines high-quality surveys and abundant administrative records; SSL is useful for transferring labels from surveys to broader unlabeled microdata.

- Sources: American Community Survey (ACS) from U.S. Census (https://www.census.gov/programs-surveys/acs), BLS public datasets.
- Modalities: tabular, geographic aggregates, time series.
- Project ideas: predict income or employment categories, semi-supervised domain adaptation from survey-labelled to administrative data, evaluate label propagation across similar regions.

6. Housing & Urban Development

   - Why: strong tabular + geospatial mix; many properties lack outcome labels (e.g., eviction, vacancy) while covariates are available.
   - Sources: HUD public datasets (https://www.hud.gov/program_offices/comm_planning), local open assessor datasets.
   - Modalities: tabular, geospatial coordinates, temporal records.
   - Project ideas: classify housing price tiers, predict risk of vacancy/eviction using semi-supervised geospatial smoothing, study label noise impact on price-category prediction.

7. Crime & Public Safety

   - Why: engaging, messy, and biased reporting; SSL helps when detailed labels exist only for a subset of incidents.
   - Sources: FBI Crime Data Explorer (https://crime-data-explorer.fr.cloud.gov/), city open data portals (Chicago, NYC, etc.).
   - Modalities: tabular, geospatial, timestamp, optional textual descriptions.
   - Project ideas: predict crime type, map high-risk zones with semi-supervised spatial propagation, co-training using spatial vs temporal/textual views.

8. Energy & Utilities

   - Why: abundant time-series consumption data with many unlabeled events; SSL leverages temporal consistency and similarity across customers/regions.
   - Sources: EIA open datasets (https://www.eia.gov/opendata/), state utility portals and smart-meter datasets.
   - Modalities: time series, tabular metadata.
   - Project ideas: classify high-demand periods, detect anomalous consumption patterns with semi-supervised pretraining, evaluate SSL for rare-event detection.

Each domain encourages students to define a clear target variable, restrict *labels* to $10 - -20\%$ for *training*, and treat the remainder as *unlabeled* for `SSL experiments`.

Deliverables

- A public **GitHub repository** named `challenge-2_<groupid>` containing:
  - *Reproducible code:* clear `Jupyter notebook(s)` or `scripts` reproducing all experiments; provide a single *best-run* notebook and separate utility modules if needed.
  - *Environment specification:* `requirements.txt` or `pyproject.toml`; a `Dockerfile` is strongly encouraged for full reproducibility.
  - Exact commands or a `Makefile` to reproduce the best reported run and to evaluate saved checkpoints; record all random seeds used for reported results.
  - *Model artifacts:* representative `trained model` files or `checkpoints`, and sample `logs` that show the pseudo-label selection process and iteration history.
- A **scientific paper** in `IEEE format`, maximum six (6) pages excluding references. The `paper` must be concise and scientific: include *abstract*, brief *related work*, *data description* (sources and preprocessing), *methods* (model and SSL details, thresholds, hyperparameters), *experimental design* (splits, seeds, number of runs), *quantitative results* with uncertainty (mean and std), *ablation analyses* if applicable, *discussion* of failure modes, and societal *implications*.
- A **short video** (7–10 minutes) uploaded to a public or unlisted link (e.g., *OneDrive*) that demonstrates the project, explains the methodology and results, and includes a brief live demo of the best model.

Reproducibility checklist

- *Fix and record* all `random seeds`.
- Provide *environment* specification (`requirements.txt` or `pyproject.toml`). `Docker` is *encouraged* but optional.
- Include `run scripts` or `Makefile` / binder instructions for *ease of evaluation*.

References and further reading

Provide canonical references in the report: the self-training literature, label propagation papers, and recent surveys on semi-supervised learning.

This challenge emphasizes principled experimentation with limited labels and real open-data problems. The pedagogical aim is to teach students how to extract signal from abundant unlabeled data while understanding the risks of incorrect pseudo-labeling.